🎈 Static Electricity Stations: Lab Sheet · ANSWER KEY

Chemistry · Unit 1A · Level 3 · goes with L1.3 Sparks, L1.5 Particle Model of Static, L1.8 Neutral Interactions

Name: _____

Question: How do charged objects interact with other charged and neutral objects?

Materials

2 balloons, paper scraps, tinsel, thin stream of water, string or tape

⚠️ Safety

- Tell staff about latex allergies.
- Keep charged balloons away from computers and outlets.
- Wipe up water spills right away.

Prediction

Predict: will a charged balloon attract or repel neutral paper? Explain using electrons.
Attract: the balloon pushes the paper's electrons away, so the near side is + (polarized).

Procedure

1. Rub a balloon on hair or wool for 10 seconds.
2. Station 1: hold it 2 cm above paper scraps. Record.
3. Station 2: hold it near tinsel. Record.
4. Station 3: hold it near a thin stream of water. Record.
5. Station 4: charge two balloons, hang them 5 cm apart. Record.
6. Move each object farther away and record if the effect gets weaker.

Data

Station	Attract or repel?	Neutral or charged object?	What happened when farther away?
1 Paper scraps	**attract**	**neutral**	**weaker**
2 Tinsel	**attract**	**neutral**	**weaker**
3 Water stream	**attract**	**neutral**	**weaker**
4 Two balloons	**repel**	**charged (both −)**	**weaker**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does a charged balloon interact with neutral objects?
It attracts them.

2. Evidence: Which stations support your claim?
Paper, tinsel and water all moved toward the balloon.

3. Reasoning: Explain using electrons, polarized, and distance.
The balloon's − charge pushes electrons to the far side; the near side is +, so it is pulled; the force is weaker farther away.

🔑 *Key written by our team from Mr. Brittle's lab sheets, please check.*